

# EXPLORING AI DISCUSSIONS ON BLUESKY

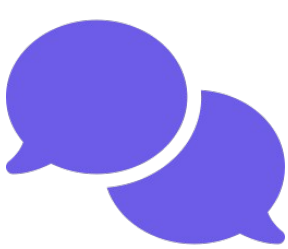
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## Topic clustering, user engagement and social network analysis

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**1** RQ1  
WHAT ARE THE MAIN TOPICS IN AI-RELATED POSTS ON BLUESKY?

**2** RQ2  
DOES USER ENGAGEMENT DIFFER ACROSS AI DISCUSSION TOPICS?

**3** RQ3  
HOW ARE AUTHORS CONNECTED THROUGH FOLLOW RELATIONSHIPS, AND WHICH ACCOUNTS ARE CENTRAL?



### DATA

Public Bluesky posts via  
atrrr, 7–8 Oct 2026



### SEARCH TERMS

artificial intelligence ·  
machine learning ·  
ChatGPT · generative AI



### TEXT PROCESSING

cleaning · stemming · TF-IDF



### ANALYSIS

k-means (RQ1) ·  
randomisation test (RQ2) ·  
follow network (RQ3)

1,168

posts collected

1,031

usable text posts

887

unique authors

5,483

TF-IDF terms

7

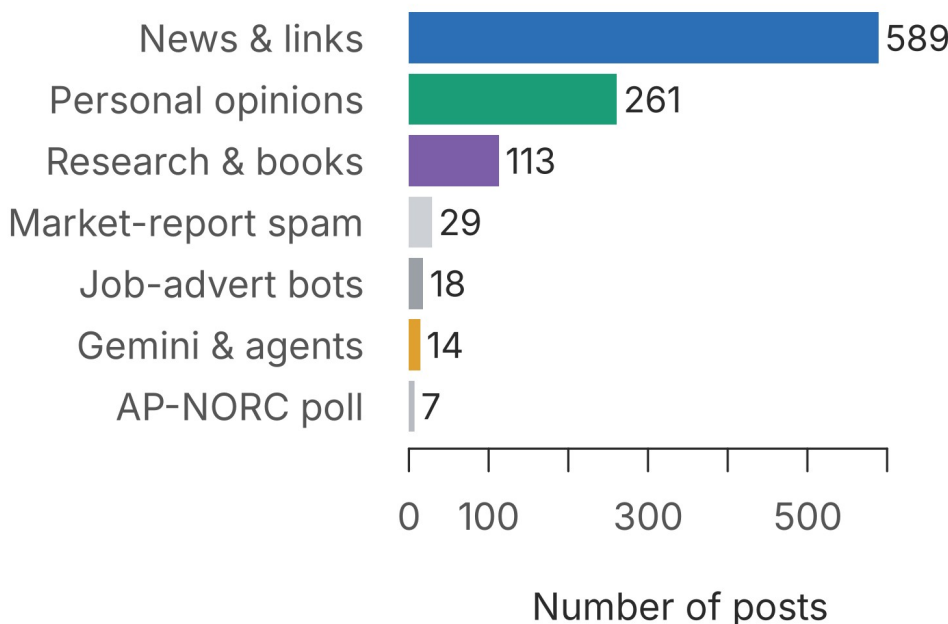
topic clusters

118,988

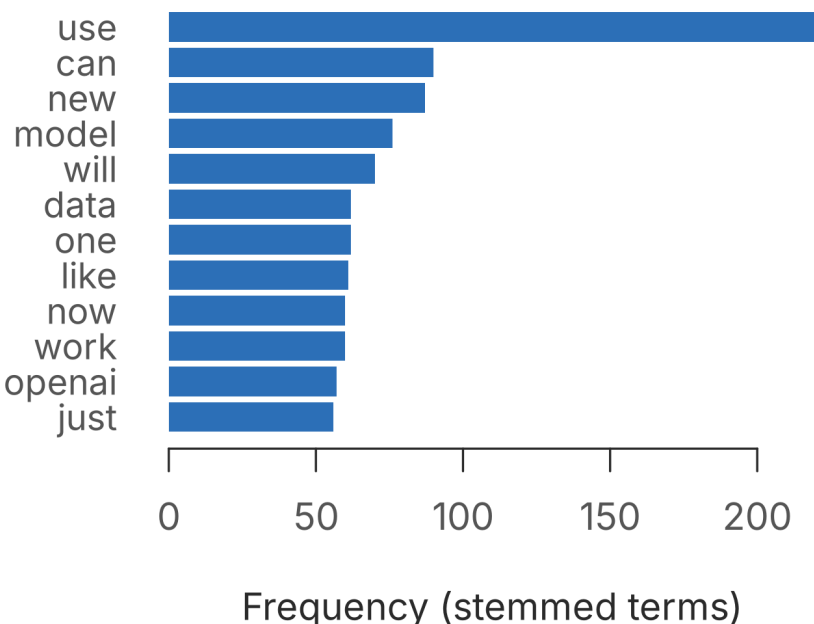
follow edges

## 1 What do people post about AI?

TF-IDF → cosine distance → MDS → k-means (k = 7)



Posts per topic cluster. Grey = automated or repeated content.



Most frequent stemmed terms: discussion of using AI in practice.

News & link sharing 57%

Personal opinions on AI 25%

These two topics dominate the sample.



3 of 7 clusters are automated or repeated content: job-advert bots, market-report spam and one poll headline posted many times.

## 2 Which topics get engagement?

Randomisation test for a difference in means (Module 10)

Engagement = likes + reposts + replies

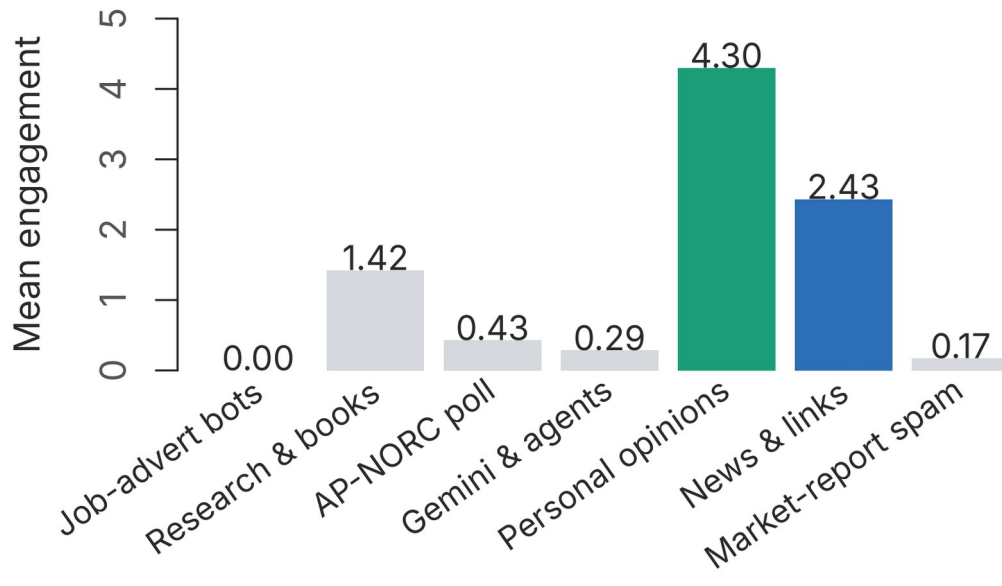
News (n = 589) vs Opinion (n = 261)

H0: equal mean engagement

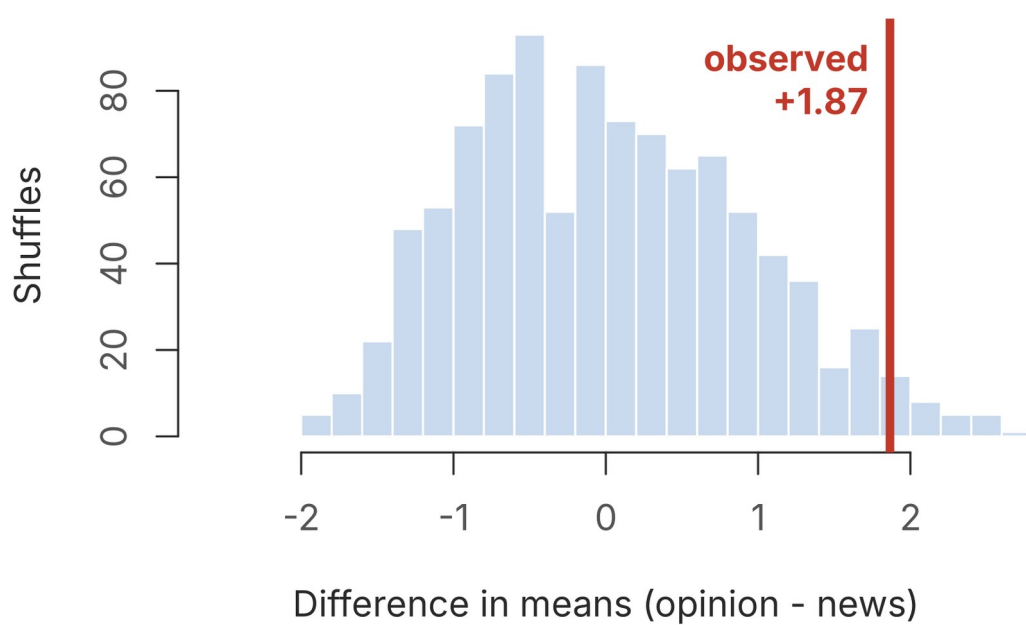
HA: means differ ( $\alpha = 0.05$ )

**p = 0.032**

1,000 shuffles · t-test p = 0.046  
Reject H0



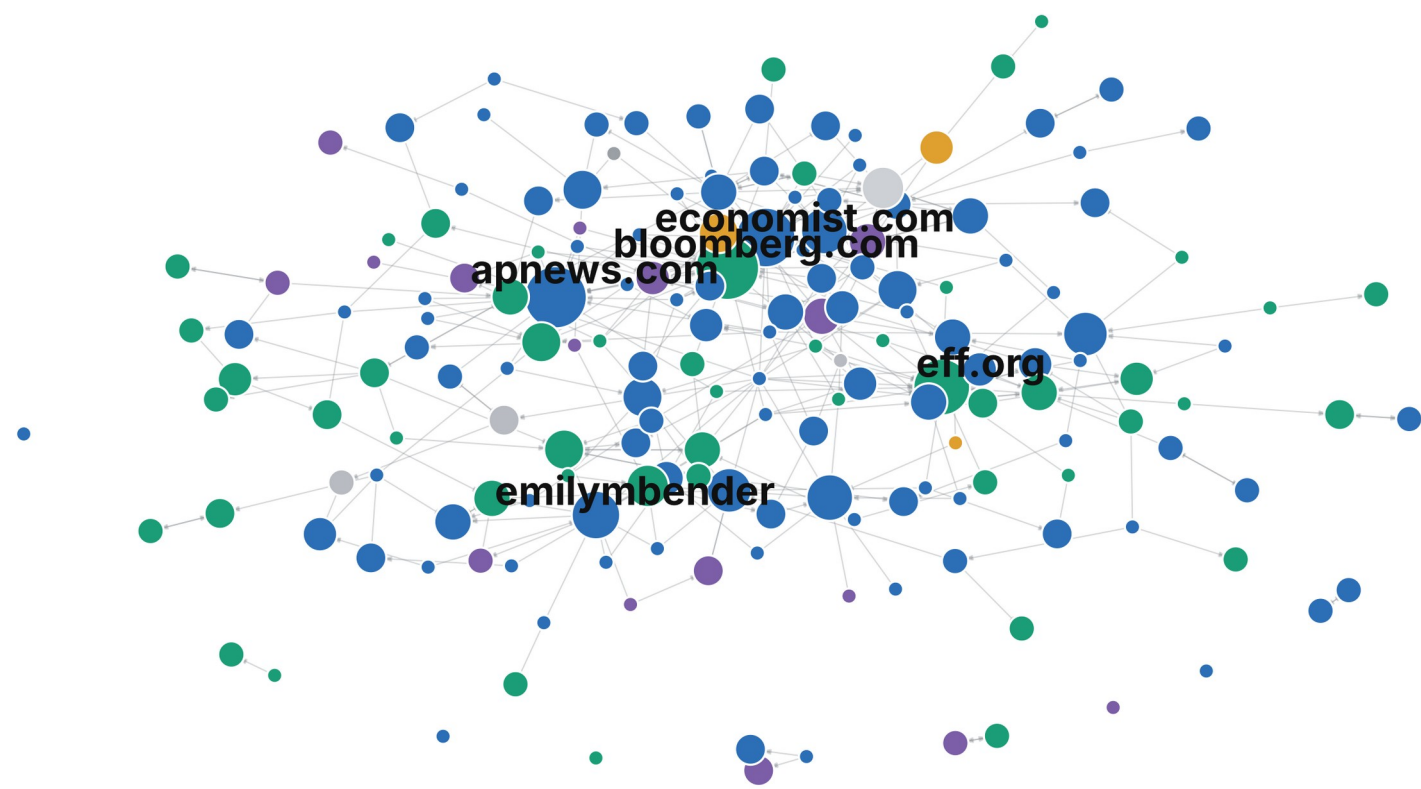
Opinion posts average 4.30 engagements vs 2.43 for news; automated clusters get almost none.



Observed difference (+1.87) lies in the tail of 1,000 label shuffles. Across all 7 clusters, ANOVA p = 0.119.

## 3 Who is connected, and who is central?

Node = author · edge A → B = A follows B (directed)



News & links Opinions Research Gemini Automated  
Authors with degree > 1 (177 nodes, 325 edges); size = in-degree.

340/803

authors with a tie

480

follow edges

0.004

density

18 weak components; the largest holds 294 of the 340 connected authors.

| # | In-degree     | PageRank      | Betweenness      |
|---|---------------|---------------|------------------|
| 1 | apnews.com    | ketanjoshi.co | vinkius-mcp-ai   |
| 2 | bloomberg.com | nposegay      | colin-fraser.net |
| 3 | eff.org       | savasavasava  | bigearthdata.ai  |
| 4 | economist.com | mattseybold   | bloomberg.com    |
| 5 | emilylbender  | apnews.com    | pauljdavies      |

**Network × engagement:** posts by connected authors get more engagement (4.12 vs 1.50,  $p < 0.001$ ). Authors mostly follow organisations (news, EFF); individual commentators rank highest on PageRank and betweenness.



### OVERALL FINDING

AI conversation on Bluesky is mostly news-sharing and personal opinion. Opinion posts, and posts by authors embedded in the follow network, receive more engagement; automated content is largely ignored. Authors mostly follow established organisations, while individual commentators hold the most central positions.

Limitations: ~1 day of keyword-search posts, not a random sample · non-English text lost in cleaning · no clear elbow for k · engagement very skewed (median 0) · max ~300 follows per author; a follow is not an interaction.